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MS-QMS Template

Design Verification Plan

Guideline: This template is applicable to ISP11 and later releases.
<Update table of content before submitting the deliverable for approval >
*<**Fill-in instructions for the user of this template***
Header:

- *Replace <Subject> by the actual subject*
- *Example deliverable: Verification Plan Shaver TopGun R4.2*

To create a document, take the following 3 rules into account:

- *Keep black text without angle brackets as is*
- *Replace black text between angle brackets by project content covering its intent*
- *Remove red text containing fill in instructions and/or examples*

All chapters and tables in this document that require information to be entered must be completed.
For content not applicable fill in "Not Applicable" or 'NA' and include justification.>

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Record ID: *DXXXXXXXXX (Rev X)*

1 Document Introduction

1.1 Purpose

This document plans (not the scheduling) the management, execution and control of the verification activities of the requirements in scope of this document.

The purpose of this plan is to identify the tasks, deliverables and resources required for Verification of '<product name, milestone>' release.

The objective of the Verification Plan document is to plan (i.e. identify the activities, methods, resources, etc.) the comprehensive verification activities that:

- Discover defects as early as possible in the product lifecycle
- Assure conformance to standards
- Increase confidence level in the reliability of the product
- Provide objective evidence that the "product was built right".

1.2 Scope

<Describe the boundaries of the applicability of the document. Scope refers to <Subject> and can be specified with more detail>

<Also, mention the events which will trigger the plan updates>

1.3 References

< This document shall include all required content. If not possible references to the additional content is allowed.

References shall refer to unique records or document stored in a document management system or published by authorities

Fill in 'Not applicable' if there are no references.

Ensure that references are mentioned in the content of the document by [REF-number]

Do not use hyperlinks as reference/Identification.>

Reference	Document Title	Document ID
REF- <1>	<Title of document>	<Document ID>
REF- <n>	<Title of document>	<Document ID>

1.4 Abbreviations and Definitions

<Use table below for definitions and abbreviations used in this template.>

Term	Definition
<abbr>	*<Abbreviation is written out>*

2 Strategy

< Give a short overview of the verification strategy and methods to be used for the scope of this document:

- *Identification of the scope of verification: e.g. the configuration of a design to be verified to meet a corresponding set of requirements, an architecture (implementing requirements adequately including interfaces), the implementation of an architecture, an integration, …*
- *Specify how the subject will be verified: e.g. technical review, code review, modeling, analysis, simulation, regression testing, smoke test, performance testing, evaluation, inspection…*
- *Consider the different types or classes of requirements that will be verified e.g. functional or feature requirements, interface requirements, reliability requirements, compliance requirements in the various domains (privacy, security, sustainability, payment card industry etc.)… Describe the verification strategy per type. Make sub sections if appropriate, for example, consider to structure the verification plan according to the outline of the requirements document…*
- *List which verification results will be reused (e.g. in case of reuse of existing parts or elements) Justify the reuse and applicability of historical verification data for the design under test*
- *Describe which verification activities will be performed on lower level of the design configuration in scope (e.g. element and/or part level)*
- *Describe which verification activities will be out-sourced.*
- *Specify the order of execution (e.g. de-risking of time and cost)*

Note: It is permitted to establish a design verification master plan, which defines the breakdown of lower level design verification plans (e.g. functions/features, interfaces, compliance domains, element verification plans e.g. software verification plan, SW item / element verification plan, unit / part verification plan,…) >

3 Test Design

<Describe the detailed verification approach in line with the outlined strategy and per type of verification, i.e. inspection, analysis, review and test.

Group verification activities with a common scope e.g. an integration, a detailed element/part design, an interface, … It is recommended to make subsections if appropriate.

Consider to group the description of the verification per requirement type e.g. functional performance, usability, reliability, regulations, claims, …. It is recommended to make subsections if appropriate.

For verification by test, list the tests with reference to their detailed protocols.

Provide per test protocol clarifying information e.g.:

- *Reference to documentation of test protocol description*
- *Information if the test protocol is e.g. new, updated, re-used, out-sourced*
- *Information about dependencies in execution of test protocols*
- *Information about the configuration to be used for the test e.g. product, element for SW also the System SW configuration the SW is verified on.*
- *Information about pass/fail or acceptance criteria e.g. within range of requirement*
- *Information about duration of a test protocol*
- *Information for risk based testing*

Tests and related Information can be listed in a table as below e.g.:

Test Protocol	*Document ID*	*Type*	*Method*	*Dependencies*	*…*

<e.g. Test Protocol >	*<Document ID>*	*<new, update, re-use, out-sourced>*	*<test/evaluation/inspection>*	*<perform after test protocol x, before test protocol y>*	

The traceability from requirements to test protocols is documented in the separate Requirements Traceability Matrix deliverable.

For complex tests with multiple test protocols, test conditions, multiple n to n relations and interactions one can detail out this cohesion, and execution order in a separate Test Design Document. Also the trace from requirement to be verified to tests can be incorporated in the Test Design Document.

For verification by inspection, review and analysis, refer to associated technical reviews or other activities and how these will be executed, e.g. inputs, decision criteria, review participants, …

Traceability is needed to inspection, analysis and technical review as well as to test protocols.>

3.1 Verification Activities Approach Overview

***Guidance**: This will describe what type of testing will be executed and how the plan will be carried through. This section will list all the requirements for the plan strategy. For test methods not used, please keep the heading and type "None" in that section.*

3.1.1 Verification Types

Guidance: Describe the different types of verification done across the lifecycle, when they are executed and how the outcome is documented.

The table shown is an example only. Add or modify the types, as applicable.

Types of Verification	When is it done?	How is it documented?
Unit Verification		
Static Verification		
Code Reviews		
Integration Testing		
System Testing		

3.2 Methods Overview

***Guidance**: Describe the candidate list of verification techniques and the rationale for selecting the ones to apply to this project. Fill in the details for each verification technique – an overview of each technique and how it will be applied on this project.*

3.2.1 Coverage Strategy

Guidance: Describe what kind of coverage at each interim development step, how much overlap with Detailed and system test, etc. I.e. describe the strategy for selecting what is verified at each phase and why.

3.2.2 Requirements Traceability Strategy

***Guidance**: Describe the method for ensuring that all originating requirements are effectively tracked forward and backward as they are decomposed and refined throughout the system's lifecycle.*

3.3 Tools Overview

Guidance: Describe the candidate tools (e.g. test automation tools, testbeds, or "gold standards", etc.) and the rationale for selecting the ones to apply to this project.
[The following tools will be using by the verification phase:

- *Quality Center – Test Management tool version xx.xx*
- *Clear Quest - CTUDT - Defects tracking common tool] version xx.xx*
- *SVN….*

All the tools used in the activity of verification needs to be validated prior to usage for its intended usage.

3.4 Test Environment Overview

***Guidance**: Describe the operating and test environment to be used in the verification process.*

3.5 Product Hazard/Risk Management Activities

***Guidance**: Describe the method for identifying requirements specifically related to product risk management so that appropriate verification focus can be applied to these. Provide reference to the RMF file*

3.6 Exploratory Testing Strategy

***Guidance:** Describe the method incorporating exploratory testing used to augment (supplement) the structured verification testing of product requirements and functional operations. Exploratory testing is intended to combine variability and randomization of parameters and steps and may include introducing stress conditions to uncover unexpected side effects. Exploratory testing cannot thus be construed to prove, by itself, requirements coverage and traceability to requirements. However, the Free Hand tests and report may optionally provide reference to Requirements on which the tests were based. These Exploratory tests are to be included in Appendix.*

3.7 Interface Testing Strategy

***Guidance:** If the intended use requires that medical device be connected to or to have interface with other medical device (s), verification shall confirm that the design output meets design inputs when so connected or interfaced. Capture the relevant details for the same in this, as applicable.*

3.8 Usability Engineering - Formative Evaluation Strategy

The purpose of the formative usability evaluation is to collect customer feedback and to identify and clarify use-related problems. The test also evaluates the effectiveness of the design solutions to reduce use errors.

The evaluation will compare performance and subjective ratings on the current design and two design concepts.>

Describe the tasks that will be evaluated, and the process for selection, characterization, and prioritization for the formative evaluation. Explain about the planning aspects for this activity.

<This is an optional section, provided formative evaluation is documented in the Usability Evaluation plan template>

4 Resources

< Describe the required verification resources regarding:

- *Locations*
- *Facilities*

- *(Qualified) tooling and test equipment*
  Note. All used tools and test equipment for verification must be qualified before verification starts
- *Human resources (e.g. special qualifications)*
- *Data recording (e.g. data storage on test site)*
- *Etc.*

>

4.1 Capital Equipment

Guidance: *Identify type, number and configuration of capital equipment required to conduct verification activities.*

4.2 Key Personnel

Guidance: *Identify key skill sets required for non-routine verification activities*

4.3 Estimates

Guidance: *Identify the non-recurring staffing/de-staffing plan by skill on a monthly basis. Information to include resource estimates for Verification Activity of each sub team and system.*

Please plan and consider the required effort for documenting the test results and the review of the test results as a part of estimation in the detailed verification planning.

5 Verification Samples

< Determine the verification sample strategy e.g. configuration, size, identification, to execute the verification and provide justification.

Consider risk based testing, statistically based testing and reuse of samples in multiple tests. If the sample is not statistically valid, explain how variation in the sample will be managed to ensure robust design, e.g. multiple test runs under different conditions.>

For Software product: The sample size is always one because the code execution does not change across units and code execution is constant for the same type of hardware.

6 Entry Criteria

<Entry criteria to be met before verification execution can start.>

Subject	Criteria
<Samples>	*<Released for verification>*
<Test Protocols>	*<Approved.>*
<Test Set-up>	*<Documented and released for verification>*
<Third Party Test Facilities>	*<Accredited and Service Level Agreement in place>*
<Human Resources>	*<Available and trained>*
<Etc.>	

Guidance: Sample entry criteria provided

- *[Established Verification Baseline: Documentation, Software and Hardware. Preliminary DMR needs to be available*
- *Availability of the approved Risk Management Matrix*
- *Established stability and reliability of the Verification Baseline*
- *Defect report count/severity*
- *Agreement on the selected defects to be resolved after RfV including agreed resolution dates*
- *Successful completion of any defined test activities prior to Verification (e.g. dry run of System Verification Specification)*
- *Availability of test environment*
- *Ensure tools used to support Verification are qualified and calibrated before use*
- *Add additional project specific entrance criteria as needed]*

Deviations to the above entrance criteria must be called out in the Deviations section of the verification report.

Entry criteria checklist as provided by the Milestone review procedure needs to be filled, reviewed and approved.

7 Exit Criteria

<Define when the verification can be ended (e.g. use exit criteria, decision meeting).>

***Guidance**: Sample exit criteria provided*

- *[Evidence of successful completion of the Verification Plan*
- *All safety and security defects reports are in an appropriate end state*
- *All defects resulting from executing the Verification Specification are either in an end state or have approved justification as to why it is acceptable to leave them open and correct them during the Validation phase]*

Exit criteria checklist as provided by the Milestone review procedure needs to be filled, reviewed and approved.

7.1 Suspension Criteria:

***Guidance**: Sample suspension criteria provided*

- *[Major project scope change(s)*
- *Stability and reliability of the system and/or test environment*
- *Equipment failure of the system and/or test environment*
- *Excessive defects influx]*

8 Acceptance Criteria

<Define the overall verification acceptance criteria (i.e. the criteria for overall success of product verification).>

9 Defect Management

<Identify the defect management process to be used for product / element / part defects, failing test set-ups, test protocol failures.>

10 Record history

Rev	Release Date	Author	Description of changes

11 Record Review and Approval

Name Author	Function Author
<Full Name>	*<Verification Lead>*

Name Reviewer(s)	Function Reviewer(s)
<Full Name>	*<System Architect/System Engineer>*
<Full Name>	*<Verification/Development Manager>*
<Full Name>	*<Quality Engineer>*
<Full Name>	*<PMO/Design Transfer>*

Name Approver	Function Approver
<Full Name>	*<Project Manager/RTE>*

12 Appendices

12.1 Management of Verification Activities

Schedules

Guidance: *Either includes herein or by reference the schedule for executing the activities identified in this plan.*

Roles and Responsibilities

Guidance: *Identify key individuals that will be responsible for planning, coordination and execution to ensure that activities identified in this plan are completed as planned. Explicitly identify the individual responsible for maintaining this plan and associated schedules.*

Status Reporting

Guidance: *The operating mechanism of the various teams and their interactions needs to be detailed in this section.*

Metrics

Guidance: *List of metrics, rationale for collecting them, mode and frequency of collection and trigger to change/rebaseline needs to be documented*

Risk Management

Guidance: *Identify potential programmatic, personnel, time schedule, overlapping tasks risks associated with the successful completion of the objectives of this plan and the risk mitigation strategy for each*

12.2 Verification Table (V-Table)

Guidance: *Identify Test, configurations and provide comments as required.*

Test Scenario	*Config #1*	*Config #2*	*Comment Section*
Scenario #1	*V*	*V*	
Scenario #2	*V*		
Scenario #3	*V*	*V*	
Scenario #4	*V*		
Scenario #5	*V*		
Scenario #6	*V*		
Scenario #7	*V*		
Scenario #8	*V*		
Automated Scenario	*V*	*V*	

12.3 Exploratory Testing

Guidance: *Identify the exploratory testing planned to evaluate specific features and functions affected by this release*

Requirement	Description of Exploratory Test for Requirement	Rationale

12.4 Expert Testing

Guidance: *Identify the expert testing planned to evaluate specific features and functions affected by this release*

Application/Feature	Expert Test to be done by	Schedule	Effort

12.5 Security Testing

Guidance: Identify the security testing planned for this release.

End of Document

Revision history

V	Change type	Change plan Id	MS-QMS content owner	Compliance approver	Change description
1	Initial version	CP_PD_0056_000026	Hernandez Perez, Carlos (310098568) (QMS Manager) at 23-Jun-2022 21:23 CEST		Initial Version. Replaces ICAP-T-030010.01 (Rev H)
2	Process improvement	CP_PD_0056_000029	Kumar, Aashish (310099170) (Training) at 22-Nov-2022 09:17 CET	Hernandez Perez, Carlos (310098568) (QMS Manager) at 03-Nov-2022 16:24 CET	Added Section 12.5 for Security testing

Publication signing manifest

Function	Approval by electronic signature	Sign-off role	Meaning
Training	22-Nov-2022 09:17 CET	Kumar, Aashish (310099170)	Approve for publication

Publication status: Approved and effective
Effective date: 22-Nov-2022 09:17 CET

Notes: Approvals are given by electronic signature
All time stamps are CET time zone in 24 hour format

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Revision history template and signing manifest version: 2